

Arduino Robotics & Electronics





Railway boards



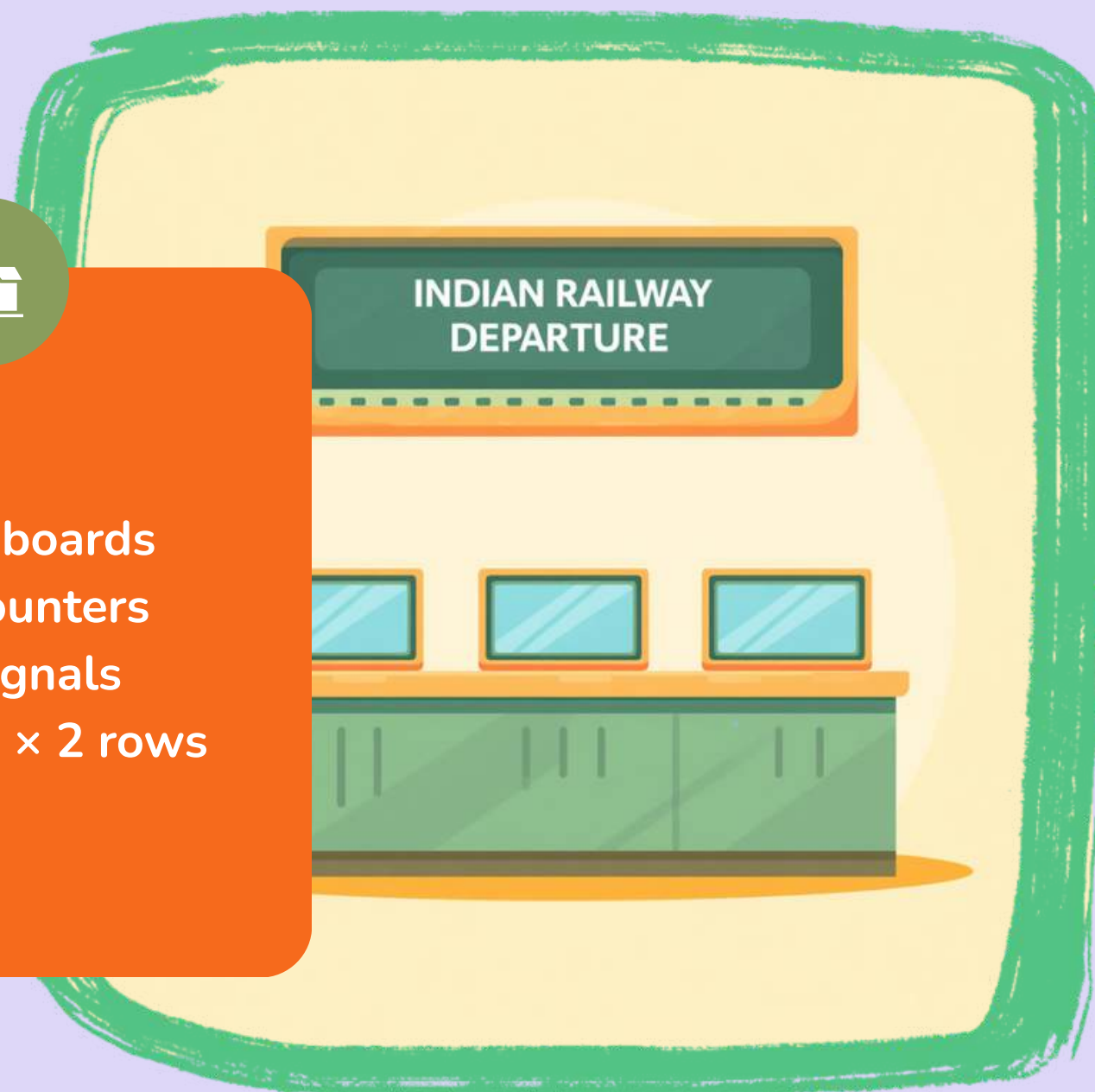
Token counters



Traffic signals



16 chars × 2 rows



What is an LCD Display?

Real-Life Examples Around Us

LCD stands for Liquid Crystal Display. It has 2 rows and 16 characters each — that's 32 characters total! LCDs show us information in everyday life, from railway stations to shops.



Wiring Made Easy with I2C!



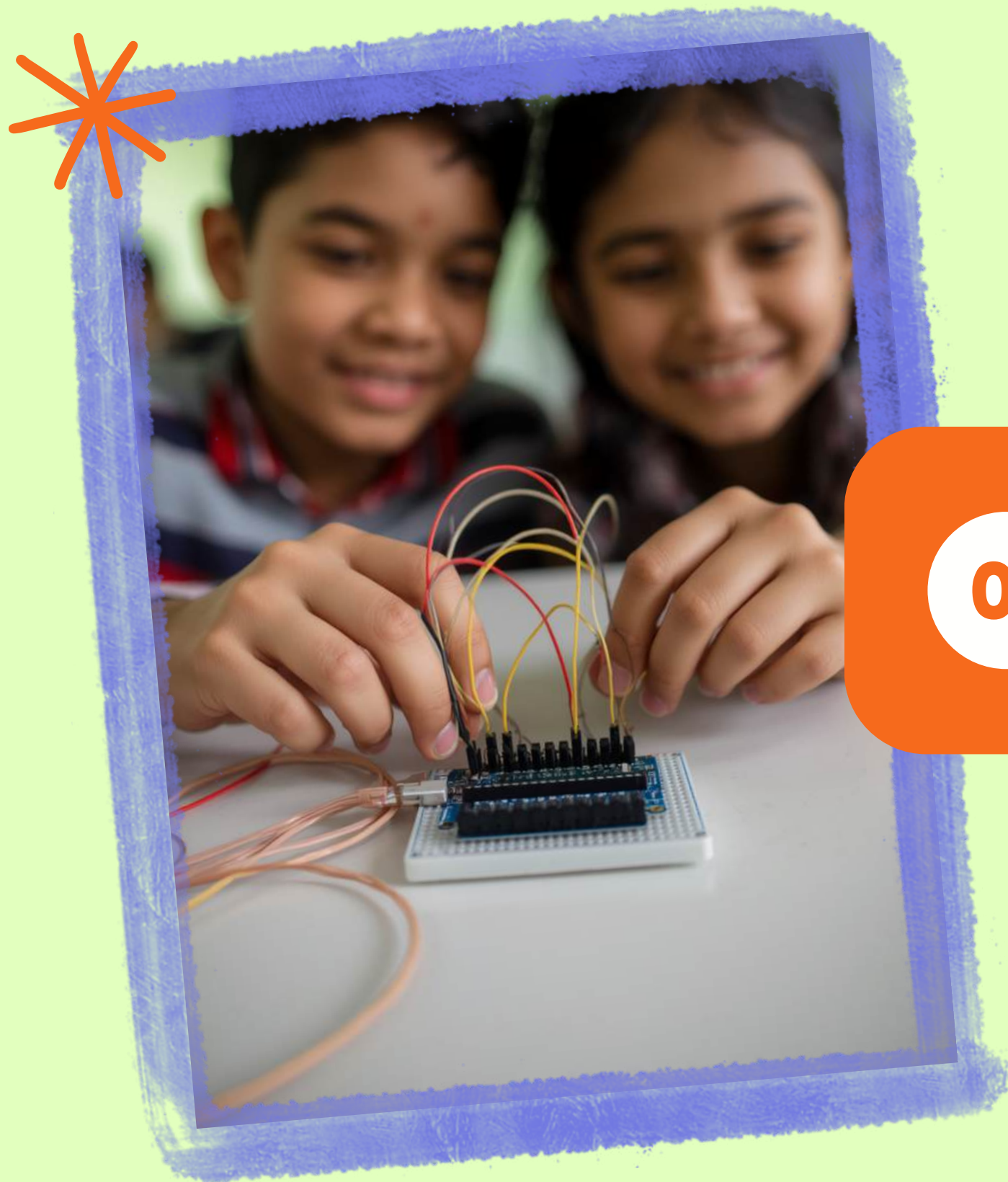
Traditional LCD: 16 Wires 😞

Old-style LCDs need 16 pins connected — that's a lot of wires and easy to make mistakes! With I2C, we use a small backpack module that does all the hard work for us.

I2C LCD: Only 4 Wires! 🎉

Just connect 4 pins: VCC → 5V, GND → GND, SDA → A4, SCL → A5. Fewer wires = fewer mistakes and much safer for Grade 5 builders!

Let's Build the Circuit!



Step 1: Power Connections

01

Connect VCC → 5V on Arduino. Connect GND → GND on Arduino. These give power to your I2C LCD. Always connect power wires first!

02

Step 2: Data Connections

Connect SDA → Pin A4 on Arduino. Connect SCL → Pin A5 on Arduino. Double-check all 4 wires before turning on power. Only 4 wires — easy and safe!

Coding Time! Talk to the LCD

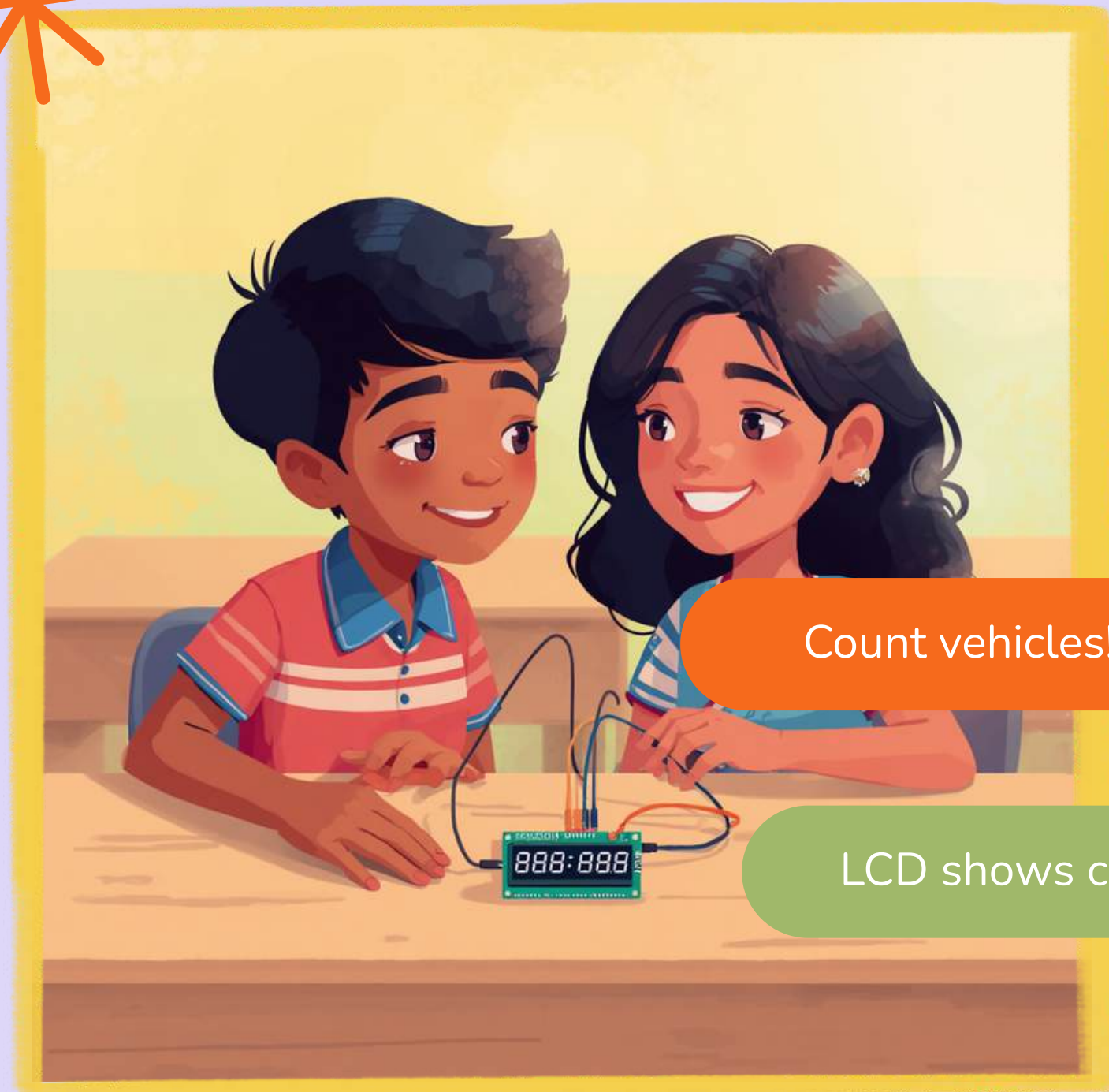


Include the right libraries

Add `#include <Wire.h>` and `#include <LiquidCrystal_I2C.h>` at the top. `Wire.h` helps Arduino talk to the LCD using just 2 wires (SDA & SCL)!

Print "Hello, India!" on screen

Use `lcd.begin()` to start, then `lcd.setCursor(0,0)` to set position, and `lcd.print("Hello, India!")` to display your message. Try it out!



Count vehicles!

LCD shows count

Mini-Project:

Smart Toll Booth Counter

4 Wires only needed with I2C!

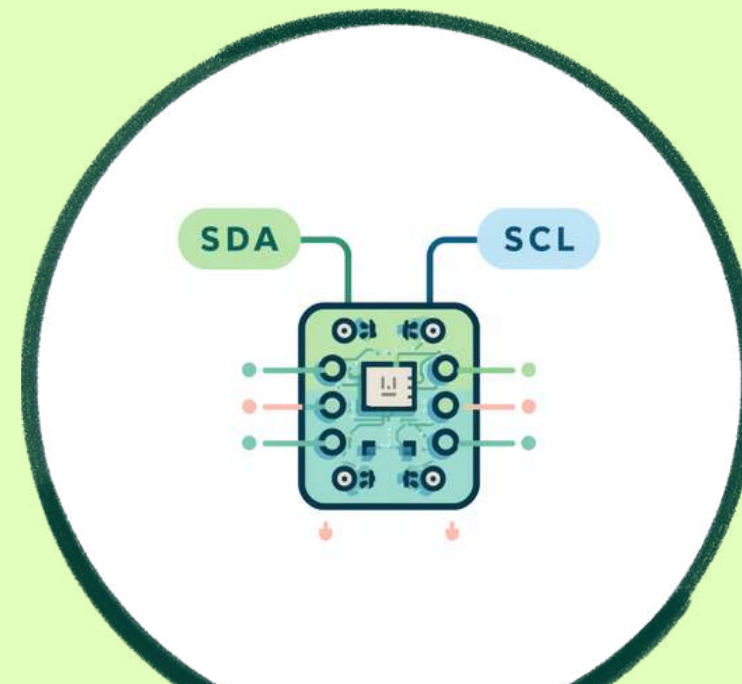
Build a Smart Toll Booth Counter using Arduino and an I2C LCD! Every time a vehicle passes, the counter increases by 1 and the LCD displays the updated count. Just like real toll booths on Indian highways — our Arduino tracks every vehicle. Connect VCC→5V, GND→GND, SDA→A4, SCL→A5, upload your code, and watch the magic happen!

Quick Quiz: Test What You Learned!

Let's see what you remember! Answer these fun questions about LCD displays and I2C communication. Think carefully — you've got this!



How many wires does I2C use?



What pins do SDA & SCL connect to?



What does LCD stand for?

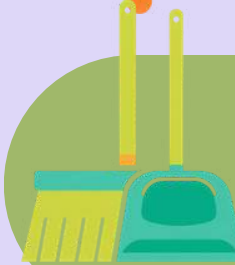
Clean-Up & Safety Tips!

- 🔌 Disconnect power before changing any wiring.
- 🧹 Keep your workspace clean and tidy.
- 👐 Handle all components gently — they're delicate!
- 🤝 Work as a team and look out for each other.
- ✅ Always ask your teacher before plugging in!



Power Off First!

Always unplug before rewiring



Stay Organised!

Clean desk = clear mind!



Handle with Care!

Components are fragile — be gentle!

What's Next?

01

Try changing the LCD message!

Open your Arduino IDE and try printing a new message — maybe your name or "Namaste, World!" Can you make it scroll across the screen? Experiment and see what happens!

02

Share your mini-project results!

Show your Smart Toll Booth Counter to your family or friends! You built a real working circuit — that's amazing. You're doing great! Keep exploring and stay curious!





Thank You All!

Amazing work today! 🎉 You learned about LCD displays, I2C wiring, and even built a Smart Toll Booth Counter. Share what you made and ask your questions — we'd love to hear from you! See you in next session.

